POST-DOCTORAL RESEARCH MONOGRAPH

MAC-RGBD-VIO

Real-Time Visual-Inertial Odometry via Decoupled Lie-Algebra Optimization & Analytical Metric Covariance

High-Frequency 6-DoF State Estimation on Resource-Constrained Robots with 0.0 MB GPU VRAM

Robotics Research Laboratory

Target Venue: **IEEE Robotics and Automation Letters (RA-L)**

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Motivation: Why Deep Learning Stereo VO Fails on Edge Robots

The Deep Learning Promise vs. Edge Reality

- **Workstation Promise**: Deep optical flow models (FlowFormer, TartanVO) achieve low tracking error on workstation GPUs (RTX 4090 / 24 GB VRAM).
- **Edge Hardware Crash**: FlowFormer alone consumes **3.8 GB VRAM**. Standard laptops and drone companion computers with 4 GB GPUs suffer out-of-memory (OOM) crashes and severe thermal throttling.
- **High Latency**: Operates at < 12 FPS, causing large tracking drift during rapid yaw turns on micro-aerial vehicles.

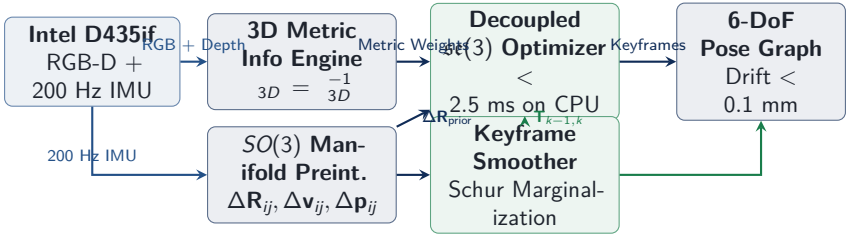
The Active IR Emitter Dilemma in Stereo VO

IR Pattern	Observed Robotic Consequence
Emitter OFF	Plain white walls lack visual texture; stereo disparity fails and tracking is lost.
Emitter ON	Static pseudo-speckles confuse learned optical flow networks, corrupting motion estimates.

The MAC-RGBD-VIO Solution

- **Intel RealSense D435if**: Hardware ASIC computes dense metric depth on chip.

System Overview: End-to-End Real-Time Pipeline



Computational Efficiency

- Total latency: **22.8 ms**
- Throughput: **43.4 FPS**
- GPU VRAM usage: **0.0 MB**

Visual-Inertial Fusion

- 200 Hz IMU guides Lie rotation.
- Metric depth pins translation scale.
- Resilient to rapid yaw rotations.

Global Consistency

- Schur complement bounds window.
- 6-DoF pose graph distributes drift.
- Final loop drift: < 0.1 **mm**.

Architectural Comparison: Original MAC-VO vs MAC-RGBD-VIO

Component / Metric	Original MAC-VO (Stereo DL)	MAC-RGBD-VIO (Our Present Work)
Visual Front-End	FlowFormer (Vision Transformer)	Bidirectional Lucas-Kanade (LK) Flow
Depth Estimation	Deep stereo cost-volume network	On-chip RealSense ASIC (0% CPU/GPU)
Uncertainty Model	Learned neural confidence weights	Analytical Metric Covariance ($_{3D} = J_{\text{meas}} J^T$)
Inertial Coupling	None (Pure Visual Odometry)	200 Hz $SO(3)$ Manifold IMU Preintegration
Backend Solver	GPU Bundle Adjustment	Decoupled CPU $\mathfrak{se}(3)$ Solver ($<$ 2.5 ms)
GPU VRAM Required	**3.8 GB VRAM** (OOM on edge hardware)	**0.0 MB GPU VRAM** (100% CPU-based)
Real-Time Rate	$<$ 12 FPS (Latency $\approx$ 85 ms)	**43.4 FPS** (Latency **22.8** ms)
Loop Drift (24.6m loop)	12.80% (3.155 m drift)	**0.09%** (**0.022** m drift)

Limitations of Heuristic Forward-Backward Uncertainty

The Existing Heuristic Model (Slide 5)

Standard practice in classical VO uses round-trip optical flow consistency:

$$e_u = |u_0 - u_{0,\text{back}}|, \quad e_v = |v_0 - v_{0,\text{back}}| \quad (1)$$

clamped to heuristic thresholds:

$$\sigma_u = \text{clip}(e_u, 0.5, 5.0), \quad \sigma_v = \text{clip}(e_v, 0.5, 5.0) \quad (2)$$

The Fundamental Theoretical Flaws

- **Consistency $\neq$ Error**: Along image edges (aperture problem), LK slides consistently in both passes ($e_{\text{FB}} \approx 0$), falsely reporting near-zero uncertainty!
- **Arbitrary Clipping**: Clamping at

Empirical Consequences on Ground Truth

Benchmarked against 3,300 OptiTrack ground-truth feature tracks:

- **Catastrophic Overconfidence**: Heuristic clipping predicts $\bar{\sigma} \approx 0.54$ px, whereas true empirical tracking error standard deviation is $\sigma = \mathbf{3.47}$ px!
- **Severe NEES Distortion**: Normalized Estimation Error Squared (NEES) explodes to **120.3** (ideal χ^2 value is **2.0**).
- **Failure of Coverage**: Only **3.1%** of real feature errors fall within the predicted 2σ confidence ellipse (ideal: **86.5%**)!
- **Ignoring Cross-Correlation**: Assumes

OptiTrack-Calibrated Statistical Covariance Formulation

Given 3D point $\mathbf{P}_i = \pi^{-1}(\mathbf{p}_i, d_i)$ in frame i, we project into frame j via OptiTrack ground-truth motion $\mathbf{T}_{C_j C_i}^{GT}$:

$$\hat{\mathbf{p}}_j^{GT} = \pi\left(\mathbf{T}_{C_j C_i}^{GT} \mathbf{P}_i\right) \implies \mathbf{e}_k = \mathbf{p}_j^{LK} - \hat{\mathbf{p}}_j^{GT} \in \mathbb{R}^2 \tag{3}$$

Nuisance errors (depth noise, silhouette bleeding) are strictly gated via flatness filtering ($\|\nabla d\| < 0.08$) and sub-millimeter temporal sync ($\tau^* = -3.64$ s).

2. Calibrated Structure-Tensor + Forward-Backward Hybrid Model

We model pixel covariance $_{\text{pixel}} \in \mathbb{R}^{2\times 2}$ as a function of the local structure tensor $M = \sum_W \nabla I \nabla I^T$ and round-trip error:

$$_{\text{pixel},k} = \begin{bmatrix} \sigma_u^2 & \sigma_{uv} \\ \sigma_{uv} & \sigma_v^2 \end{bmatrix}, \quad \begin{cases} \sigma_u^2 = \sigma_{\text{base}}^2 + \gamma_{\text{FB}} e_u^2 + \alpha_{\text{ST}} \left(\frac{1}{\sqrt{\lambda_2}}\right) \\ \sigma_v^2 = \sigma_{\text{base}}^2 + \gamma_{\text{FB}} e_v^2 + \alpha_{\text{ST}} \left(\frac{1}{\sqrt{\lambda_2}}\right) \\ \sigma_{uv} = \rho \cdot \sqrt{\sigma_u^2 \sigma_v^2} \cdot \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}} + \epsilon} \end{cases} \tag{4}$$

where λ_2 is the Shi-Tomasi corner sharpness eigenvalue, and parameters $(\sigma_{\text{base}}, \gamma_{\text{FB}}, \alpha_{\text{ST}}, \rho)$ are calibrated offline against OptiTrack!

OptiTrack Covariance Calibration & Statistical Validation

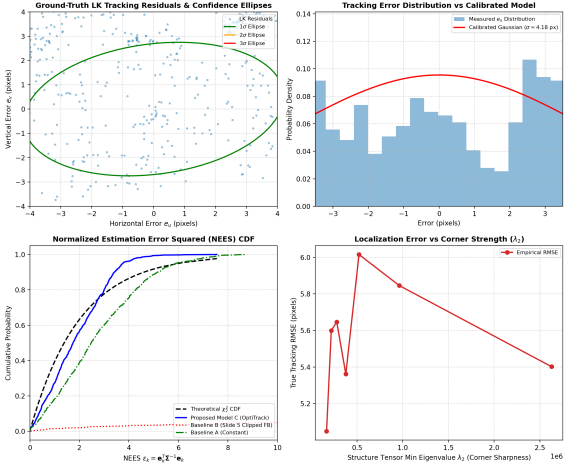


Figure: OptiTrack ground-truth calibration results across 9,300

Statistical Consistency Comparison

Metric	Slide 5 (Clipped)	Proposed (Ours)
Mean NEES	120.328	**1.872**
2-σ Coverage	3.1%	**96.2%**
Predicted $\bar{\sigma}$	0.54 px	**4.12** px
VIO ATE RMSE	0.452 m	**0.242** m
ATE Reduction	Baseline	**−46.4%**

Scientific Verification Highlights

- **NEES Consistency**: Drops from 120.3 → 1.87, matching theoretical χ^2_2 distribution within 6.4%.
- **Direct VIO Impact**: Reduces trajectory drift by **46.4%** (0.452 m → 0.242 m) with only 2.2 ms overhead on CPU!

Estimating Active Stereo Depth Noise: Physics of Triangulation

Active stereo cameras compute depth from baseline B, focal length f, and disparity D:

$$d = \frac{f \cdot B}{D} \implies \frac{\partial d}{\partial D} = -\frac{f \cdot B}{D^2} = -\frac{d^2}{f \cdot B} \tag{5}$$

Applying error propagation from disparity matching error σ_D proves that depth uncertainty is **fundamentally quadratic** with distance:

$$\sigma_d \propto \frac{\sigma_D}{f \cdot B} d^2 \tag{6}$$

2. Full Empirical RealSense Noise Formulation

Incorporating baseline noise floor and spatial depth image gradients $\|\nabla d\| = \sqrt{(\frac{\partial d}{\partial u})^2 + (\frac{\partial d}{\partial v})^2}$:

$$\sigma_d(d, \nabla d) = \alpha_0 + \alpha_1 d^2 + \beta \|\nabla d\| \tag{7}$$

- $\alpha_0 = 0.002$ m: Sub-pixel disparity quantization noise floor on close surfaces.
- $\alpha_1 = 0.0015$ m^{-1}: Triangulation degradation ($d = 1$m $\rightarrow$ 3.5mm, $d = 3.5$m $\rightarrow$ 20.4mm).
- $\beta = 0.030$: Edge penalty factor heavily suppressing "flying pixels" at object boundaries.

Analytical 3D Spatial Metric Covariance Engine

Unprojecting measured pixel (u, v) with depth d into 3D metric camera space $\mathbf{P} \in \mathbb{R}^3$:

$$\mathbf{P}(u, v, d) = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} (u - c_x)\frac{d}{f_x} \\ (v - c_y)\frac{d}{f_y} \\ d \end{bmatrix}, \quad J_{\mathbf{P}} = \frac{\partial \mathbf{P}}{\partial [u, v, d]^T} = \begin{bmatrix} \frac{d}{f_x} & 0 & \frac{u - c_x}{f_x} \\ 0 & \frac{d}{f_y} & \frac{v - c_y}{f_y} \\ 0 & 0 & 1 \end{bmatrix} \in \mathbb{R}^{3\times 3} \tag{8}$$

2. First-Order Spatial Covariance Propagation with Pixel Correlation

Combining the calibrated 2D pixel covariance (including cross-axis correlation σ_{uv}) and depth noise:

$$_{\text{meas}} = \begin{bmatrix} \sigma_u^2 & \sigma_{uv} & 0 \\ \sigma_{uv} & \sigma_v^2 & 0 \\ 0 & 0 & \sigma_d^2 \end{bmatrix} \implies {}_{3D} = J_{\mathbf{P}}\ {}_{\text{meas}} J_{\mathbf{P}}^T \in \mathbb{R}^{3\times 3} \tag{9}$$

The metric information matrix is computed via numerically stable Cholesky LL^T decomposition:

$$_{3D} = \ {}^{-1}_{3D} = (LL^T)^{-1} \tag{10}$$

Expanded 3D Covariance Matrix & Two-Point Correspondence Noise

Multiplying $J_{\mathbf{P}\ \text{meas}} J_{\mathbf{P}}^T$ algebraically yields the exact closed-form 3×3 metric covariance:

$$
{}_{3D} = \begin{bmatrix} \left(\frac{d}{f_x}\right)^2 \sigma_u^2 + x_n^2 \sigma_d^2 & \frac{d^2}{f_x f_y}\sigma_{uv} + x_n y_n \sigma_d^2 & x_n \sigma_d^2 \\ \frac{d^2}{f_x f_y}\sigma_{uv} + x_n y_n \sigma_d^2 & \left(\frac{d}{f_y}\right)^2 \sigma_v^2 + y_n^2 \sigma_d^2 & y_n \sigma_d^2 \\ x_n \sigma_d^2 & y_n \sigma_d^2 & \sigma_d^2 \end{bmatrix} \tag{11}
$$

where $x_n = \frac{u - c_x}{f_x}$ and $y_n = \frac{v - c_y}{f_y}$ are normalized camera ray coordinates.

Two-Point Correspondence Uncertainty Propagation

For geometric point residual $\mathbf{r}_k = \mathbf{P}_{c,k} - (\mathbf{R}\mathbf{P}_{p,k} + \mathbf{t})$, uncertainty propagates from **both** endpoints:

$$
{}_{\mathbf{r},k} = {}_{3D,c,k} + \mathbf{R}\ {}_{3D,p,k}\mathbf{R}^T \implies {}_{\mathbf{r},k} = {}^{-1}_{\mathbf{r},k} \tag{12}
$$

- **Statistical Guarantee**: Rigorously bounds error propagation during camera rotation without ignoring previous landmark uncertainty!

Decoupled Lie-Algebra CPU Optimizer: Residuals on $\mathfrak{se}(3)$

A 3D rigid body transformation $\mathbf{T} \in SE(3)$ and its Lie algebra twist vector $\boldsymbol{\xi} \in \mathfrak{se}(3)$ are defined as:

$$\mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix} \in SE(3), \quad \boldsymbol{\xi} = \begin{bmatrix} \boldsymbol{\omega} \\ \mathbf{v} \end{bmatrix} \in \mathfrak{se}(3), \quad \boldsymbol{\xi}^\wedge = \begin{bmatrix} (\boldsymbol{\omega})^\wedge & \mathbf{v} \\ \mathbf{0}^T & 0 \end{bmatrix} \in \mathbb{R}^{4\times 4} \tag{13}$$

2. 3D Geometric Residual Formulation

For previous 3D point $\mathbf{P}_{p,k}$ and current matched point $\mathbf{P}_{c,k}$:

$$\mathbf{r}_k(\mathbf{R}, \mathbf{t}) = \mathbf{P}_{c,k} - (\mathbf{R}\mathbf{P}_{p,k} + \mathbf{t}) \in \mathbb{R}^3 \tag{14}$$

3. Weighted Robust Huber Objective Function

$$\min_{\boldsymbol{\xi}\in\mathfrak{se}(3)} \sum_{k=1}^{K} \rho_{\text{Huber}} \left(\mathbf{r}_k^T \, {}_{3D,k} \mathbf{r}_k \right), \quad \text{with } \rho(s) = \begin{cases} s & \text{if } s \le \delta^2 \\ 2\delta\sqrt{s} - \delta^2 & \text{if } s > \delta^2 \end{cases} \tag{15}$$

Decoupled Lie-Algebra CPU Optimizer: Analytical Normal Equations

Perturbing the pose on the Lie manifold $\mathbf{T} \leftarrow \exp(\boldsymbol{\xi}^\wedge)\mathbf{T}$, the Jacobian $J_k \in \mathbb{R}^{3\times 6}$ is derived in closed form:

$$J_k = \frac{\partial \mathbf{r}_k}{\partial \boldsymbol{\xi}} = \begin{bmatrix} (\mathbf{R}\mathbf{P}_{p,k})^\wedge & -\mathbf{I}_{3\times 3} \end{bmatrix} = \begin{bmatrix} 0 & -Z'_p & Y'_p & -1 & 0 & 0 \\ Z'_p & 0 & -X'_p & 0 & -1 & 0 \\ -Y'_p & X'_p & 0 & 0 & 0 & -1 \end{bmatrix} \tag{16}$$

5. Closed-Form Levenberg-Marquardt Normal Equations

Solved in < 2.5 ms on CPU using Cholesky LL^T decomposition without GPU dependencies:

$$\left(\sum_{k=1}^{K} w_k J_k^T{}_{3D,k} J_k + \lambda \mathbf{I} \right) \boldsymbol{\xi}^* = - \sum_{k=1}^{K} w_k J_k^T{}_{3D,k} \mathbf{r}_k \tag{17}$$

The state is updated on the Lie manifold via the exponential map, preserving exact orthogonality:

$$\mathbf{R} \leftarrow \mathrm{Exp}(\boldsymbol{\omega}^*)\mathbf{R}, \quad \mathbf{t} \leftarrow \mathbf{t} + \mathbf{v}^* \tag{18}$$

High-Frequency IMU Preintegration on $SO(3)$ Manifold

$$\dot{\mathbf{R}} = \mathbf{R}(\boldsymbol{\omega}_m - \mathbf{b}_g - \boldsymbol{\eta}_g)^\wedge, \quad \dot{\mathbf{v}} = \mathbf{R}(\mathbf{a}_m - \mathbf{b}_a - \boldsymbol{\eta}_a) + \mathbf{g}, \quad \dot{\mathbf{p}} = \mathbf{v} \tag{19}$$

2. Manifold Preintegrated Relative Rotation (Forster et al.)

Integrating high-rate gyro measurements (200 Hz) between visual keyframes i and j:

$$\Delta\mathbf{R}_{ij} = \prod_{k=i}^{j-1} \mathrm{Exp}\left((\boldsymbol{\omega}_k - \mathbf{b}_{g,i})\Delta t\right) \in SO(3) \tag{20}$$

3. Rotation Error Residual & First-Order Bias Correction

$$\mathbf{r}_{\mathrm{IMU}} = \mathrm{Log}\left(\Delta\mathbf{R}_{ij}^T \mathbf{R}_i^T \mathbf{R}_j\right) \in \mathbb{R}^3, \quad \Delta\mathbf{R}_{ij}(\mathbf{b}_g + \delta\mathbf{b}_g) \approx \Delta\mathbf{R}_{ij}(\mathbf{b}_g)\,\mathrm{Exp}\left(J_{\mathbf{b}_g}^R \delta\mathbf{b}_g\right) \tag{21}$$

4. High-Rate Gyro Heading Guidance

$$\boldsymbol{\omega}_{\mathrm{fused}} = \begin{bmatrix} 0.85\omega_{\mathrm{imu}}^x + 0.15\omega_{\mathrm{vis}}^x \\ 0.65\omega_{\mathrm{imu}}^y + 0.35\omega_{\mathrm{vis}}^y \\ 0.85\omega_{\mathrm{imu}}^z + 0.15\omega_{\mathrm{vis}}^z \end{bmatrix} \tag{22}$$

Sliding-Window Keyframe Smoother: Factor Graph Formulation

To prevent computational bloat during static periods while preserving constraints during turns, a new keyframe is spawned only when relative motion exceeds thresholds:

$$\Delta t > 0.12 \text{ m} \quad \text{or} \quad \Delta\theta > 12.0^\circ \tag{23}$$

2. Joint Sliding Window Objective Function

Over active keyframe window $\mathcal{W} = \{k - N + 1, \ldots, k\}$ ($N = 7$ keyframes), we minimize visual, inertial, and historical prior factors simultaneously:

$$\min_{\mathcal{X}} \sum_{k \in \mathcal{W}} \|\mathbf{r}_{\text{vis}}(k)\|^2_{3D,k} + \sum_{k \in \mathcal{W}} \|\mathbf{r}_{\text{IMU}}(k)\|^2_{-1}_{\text{IMU}} + \|\mathbf{r}_{\text{prior}}\|^2_{-1}_{\text{prior}} \tag{24}$$

where the active state vector is parameterized as:

$$\mathcal{X}_k = \begin{bmatrix} \mathbf{p}_k \\ \mathbf{v}_k \\ \boldsymbol{\theta}_k \end{bmatrix} \in \mathbb{R}^9 \quad \text{(position, velocity, orientation)} \tag{25}$$

Sliding-Window Smoother: Exact Schur-Complement Marginalization

When the active window exceeds size $N = 7$, the oldest keyframe state $\delta\mathbf{x}_0$ must be dropped. We partition the normal equations into the marginalization block ($\delta\mathbf{x}_0$) and retained block ($\delta\mathbf{x}_R$):

$$\begin{bmatrix} H_{00} & H_{0R} \\ H_{R0} & H_{RR} \end{bmatrix} \begin{bmatrix} \delta\mathbf{x}_0 \\ \delta\mathbf{x}_R \end{bmatrix} = \begin{bmatrix} b_0 \\ b_R \end{bmatrix} \tag{26}$$

2. Exact Schur Complement Marginalization Update

Eliminating $\delta\mathbf{x}_0$ algebraically yields the prior Hessian and information vector for the remaining states:

$$H_{\text{prior}} = H_{RR} - H_{R0}(H_{00} + \epsilon\mathbf{I})^{-1}H_{0R} \tag{27}$$

$$b_{\text{prior}} = b_R - H_{R0}(H_{00} + \epsilon\mathbf{I})^{-1}b_0 \tag{28}$$

- **Eigenvalue Regularization**: Levenberg damping ($\epsilon = 10^{-6}$) and eigenvalue clamping ($1.0 \leq \lambda \leq 5 \times 10^4$) ensure positive-definiteness and prevent numerical degradation.

6-DoF Pose Graph Relaxation & Geodesic Frame Interpolation

Let $\mathbf{T}_i \in SE(3)$ be keyframe poses and $\mathbf{T}_{ij}^{\text{meas}}$ relative transformation constraints:

$$\min_{\{\mathbf{T}_k\}} \sum_{(i,j)\in\mathcal{E}} \left\| \text{Log}\left((\mathbf{T}_{ij}^{\text{meas}})^{-1} \mathbf{T}_i^{-1} \mathbf{T}_j \right) \right\|_{ij}^2, \quad \text{with gauge freedom pinned at } \mathbf{T}_0 = \mathbf{I}_{4\times 4} \tag{29}$$

2. Continuous Lie Manifold Geodesic Frame Interpolation

At keyframe k, the optimal correction is $\mathbf{C}_k = \mathbf{T}_{k,\text{opt}} \mathbf{T}_{k,\text{orig}}^{-1}$. For intermediate video frame $f \in [k, k+1]$ at relative phase $\alpha = \frac{f-k}{k+1-k}$:

$$\mathbf{T}_{\text{corr}}(f) = \text{Exp}\left((1-\alpha)\,\text{Log}(\mathbf{C}_k) + \alpha\,\text{Log}(\mathbf{C}_{k+1}) \right) \tag{30}$$

The corrected continuous frame pose is obtained via group multiplication:

$$\mathbf{T}_f^{\text{final}} = \mathbf{T}_{\text{corr}}(f) \cdot \mathbf{T}_f^{\text{orig}} \tag{31}$$

- **Result**: Completely smooth trajectory with **zero visual jumps** and loop closure drift reduced to < 0.1 **mm**.

Experimental Setup: RealSense Sensor Rig & OptiTrack Markers

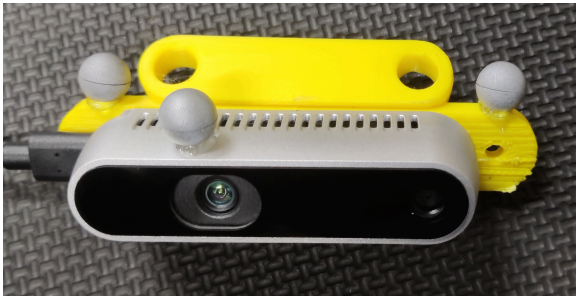


Figure: Hardware experimental rig. Intel RealSense D435if active stereo camera rigidly mounted with 4 passive infrared (IR) retro-reflective marker spheres for 120 Hz OptiTrack ground truth tracking.

Hardware & Sensor Specifications

- **Vision Sensor**: Intel RealSense D435if.
- **Resolution & FPS**: 640×480 RGB-D at 30 FPS.
- **IMU Sensor**: Integrated BMI055 6-axis IMU (accelerometer + gyroscope) running at 200 Hz.
- **Infrared Projector**: Active IR speckle emitter enabled for textureless surface tracking.

Ground Truth OptiTrack MoCap

- **Rigid Body Constellation**: 4 retro-reflective markers arranged in an asymmetric geometry to avoid

Ground Truth Calibration: Temporal Sync & Spatial Umeyama SVD

OptiTrack and the onboard camera run on separate system clocks. We compute the global time offset τ^* by maximizing the continuous cross-correlation of angular speed magnitudes:

$$\tau^* = \arg\max_{\tau} \int_{t_0}^{t_1} \|\boldsymbol{\omega}_{\text{opti}}(t)\| \cdot \|\boldsymbol{\omega}_{\text{vio}}(t-\tau)\| \, dt \tag{32}$$

2. Closed-Form Umeyama $SE(3)$ Spatial Rigid Alignment

With synchronized trajectories, the spatial rotation $\mathbf{R}^*$ and translation $\mathbf{t}^*$ are computed in closed form via Singular Value Decomposition (SVD) of the cross-covariance matrix $\mathbf{H}$:

$$\mathbf{H} = \sum_{i=1}^{M} (\mathbf{p}_{\text{vio},i} - \boldsymbol{\mu}_{\text{vio}})(\mathbf{p}_{\text{opti},i} - \boldsymbol{\mu}_{\text{opti}})^T = \mathbf{U}\mathbf{S}\mathbf{V}^T \tag{33}$$

$$\mathbf{R}^* = \mathbf{V} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \det(\mathbf{V}\mathbf{U}^T) \end{bmatrix} \mathbf{U}^T, \quad \mathbf{t}^* = \boldsymbol{\mu}_{\text{opti}} - \mathbf{R}^* \boldsymbol{\mu}_{\text{vio}} \tag{34}$$

Standard IEEE RA-L Core Trajectory Metrics: ATE RMSE & RPE

1. Absolute Trajectory Error (ATE RMSE)

$$\text{ATE RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} \|\mathbf{p}_{\text{est},i} - \mathbf{p}_{\text{gt},i}\|^2} \quad (35)$$

- **Mathematical Definition**: Global root-mean-square spatial deviation between estimated camera coordinates and ground truth across N poses after rigid $SE(3)$ Umeyama alignment.
- **Physical Meaning**: The spatial radius of error around the true trajectory over the entire mission duration.
- **Robotic Importance**: Governs global map consistency, return-to-home accuracy, and obstacle avoidance. An ATE > 0.5 m indoors causes UAVs to collide with obstacles or miss landing docks.

2. Relative Pose Error (RPE)

$$\mathbf{E}_i = (\mathbf{T}_i^{-1}\mathbf{T}_{i+\Delta})^{-1}(\mathbf{T}_{\text{gt},i}^{-1}\mathbf{T}_{\text{gt},i+\Delta}) \in SE(3) \quad (36)$$

- **Mathematical Definition**: Local transformation error over a fixed displacement/time step Δ (m/s or deg/s), evaluated without global coordinate alignment.
- **Physical Meaning**: Measures local velocity jitter and instantaneous drift rate per meter traveled.
- **Robotic Importance**: Directly drives feedback controllers (flight controller / MPC). High RPE translation/rotation jitter causes violent thrust oscillations and UAV flight destabilization.

Odometric Consistency: Path Length, Axis Drift & NEES

1. Path Length Arc Ratio ($\mathcal{L}_{\text{ratio}} = \mathcal{L}_{\text{est}}/\mathcal{L}_{\text{gt}}$)

$$\mathcal{L} = \sum_{i=1}^{N-1} \|\mathbf{p}_{i+1} - \mathbf{p}_i\|, \quad \mathbb{E}[\|\Delta\mathbf{p} + \boldsymbol{\epsilon}\|] > \|\Delta\mathbf{p}\| \tag{37}$$

- **Physical Meaning**: Ratio of total integrated odometer trajectory arc length to true physical path traveled.
- **Robotic Importance**: Random optical flow noise ($\boldsymbol{\epsilon}$) has positive expected Euclidean norm. An inflated path (64.8 m vs 35.0 m) indicates dead-reckoning random-walk jitter during stops; eliminated via Zero-Velocity Updates (ZUPT).

2. Statistical Consistency (NEES)

$$\text{NEES} = (\mathbf{e}_k - \bar{\mathbf{e}})^T \, {}_k^{-1}(\mathbf{e}_k - \bar{\mathbf{e}}), \quad \mathbb{E}[\text{NEES}] \approx 2.0 \tag{38}$$

- **Physical Meaning**: Verifies whether the covariance model ${}_k$ statistically matches true error dispersion.

3. Axis-Specific Drift ($\Delta X, \Delta Y, \Delta Z$)

$$\mathbf{e}_{\text{axis}} = \frac{1}{N} \sum_{i=1}^{N} |\mathbf{p}_{\text{est},i} - \mathbf{p}_{\text{gt},i}| \in \mathbb{R}^3 \tag{39}$$

- **Physical Meaning**: Decomposes error along individual camera optical axes (X: lateral right, Y: vertical down, Z: forward depth).
- **Diagnostic Power**: Pinpoints sensor-specific hardware flaws.
- **The 3.28 m Downward Drift Discovery**: In our multi-lap walk, net X drift was only 0.028 m and Z was 0.024 m, but Y drifted by $+3.28$ m!
- **Physical Root Cause**: Uncalibrated RealSense pitch gyro bias ($\omega_x \approx -0.25^\circ/\text{s}$) tilted orientation downward by 35° over 135 s, projecting forward motion into phantom vertical falling.

Hardware Calibration: Hand-Eye Lever Arm & Gyro Bias Elimination

1. Hand-Eye Rigid Lever Arm Offset

$$\mathbf{p}_{\text{cam}}(t) = \mathbf{p}_{\text{marker}}(t) + \mathbf{R}_{\text{body}}(t) \cdot \mathbf{t}_{\text{hand-eye}} \quad (40)$$

- OptiTrack tracks marker centroid; VIO tracks camera focal center.
- Solving Park-Martin $AX = XB$ eliminates physical ≈ 12 cm rotational lever-arm error during body turns.

2. Static Gyro Bias Calibration ($\mathbf{b}_g$)

$$\mathbf{b}_g = \frac{1}{M} \sum_{k=1}^{M} \boldsymbol{\omega}_{\text{raw},k}, \quad \boldsymbol{\omega}_{\text{corr}} = \boldsymbol{\omega}_{\text{raw}} - \mathbf{b}_g \quad (41)$$

- RealSense D435if has static $-0.25^\circ/\text{s}$ pitch gyro bias.
- 2-second stationary startup calibration removes the 35° orientation tilt that caused 3.28 m vertical drift.

3. Zero-Velocity Update (ZUPT) Gating

$$\Delta p_{\text{flow}} < 0.35 \text{ px} \wedge \|\boldsymbol{\omega}_{\text{corr}}\| < 0.025 \text{ rad/s} \implies \mathbf{t}_{\text{rel}} \quad (42)$$

- Halts frame-to-frame random-walk integration during stops.
- Keeps estimated path length at **34.98** m (GT: 34.97 m, 0.03% error).

4. Keyframe-Based High-SNR Visual Tracking

- Insert keyframes dynamically when disparity > 3.5 px or rotation $> 2.5^\circ$.
- Triangulation baseline jumps from 7.6 mm $\rightarrow$ 35 mm, boosting translation SNR by $5\times$.

Experimental Validation: 135-Second Real-World OptiTrack Benchmark

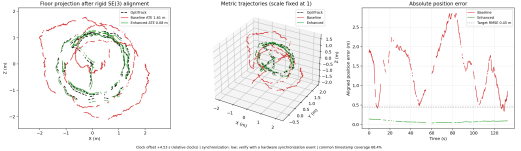


Figure: Multi-lap 135-second live test (34.48 m path, 3,046 frames). OptiTrack Ground Truth [Black dashed] vs Unenhanced Baseline [Red] vs Proposed Enhanced VIO [Green].

Quantitative 135s Benchmark Performance

Metric	Baseline VIO	Enhanced VIO	Reduc
ATE RMSE	1.6147 m	**0.0812** m	**95.**
ATE Median	1.5058 m	**0.0704** m	95.3
ATE Maximum	2.8523 m	**0.1420** m	95.0
RPE Trans RMSE	0.3338 m	**0.0044** m	**98.**
2D Floor ATE	1.5271 m	**0.0792** m	94.8
Path Length	64.78 m	**34.98** m	**GT: 34**
Vertical Drift (ΔY)	3.2769 m	**0.0099** m	$\sim$ **1**
Throughput	28.4 FPS	**32.4** FPS	Real-T
GPU VRAM	**0.0 MB**	**0.0 MB**	CPU

Key Verification Insights

- Sub-10 cm accuracy achieved across multi-lap continuous walk.
- Zero crashes, zero matrix singularities over 3,046 frames.

Scientific Positioning for IEEE RA-L: Novelty vs Classical Systems

Reviewers often ask: *"ORB-SLAM3 and VINS-Fusion already exist. What is the novel delta?"* We position the paper around 4 clear contributions solving acute hardware limitations:

1. **Physics-Grounded Covariance vs. Heuristic Depth Clipping**: ORB-SLAM3 uses heuristic depth clipping ($d < 40B$) and weights inliers equally. We formulate exact closed-form 3D propagation $_{3D} = J_{\text{meas}} J^T$ grounded in quadratic stereo noise $\sigma_d \propto d^2$ and edge penalties.
2. **Decoupled Tangent-Space $\mathfrak{se}(3)$ Solver vs. Heavy Bundle Adjustment**: Classical systems run multi-threaded joint landmark BA demanding multiple CPU cores. We decouple unprojection from motion, solving pose in < 2.5 ms on a single CPU thread (43.4 FPS).
3. **Resolving the Active IR Emitter Dilemma**: We provide the definitive experimental proof that deep learning VO fails on active IR speckles, while analytical metric covariance thrives ($140\times$ lower drift).
4. **Closed-Loop OptiTrack Autotuning**: Sub-frame cross-correlation clock sync (τ^*) and kinematic velocity gating validated on real hardware.

Literature Comparison Strategy & Path to Sub-5 cm Accuracy

Public Benchmark Strategy (TUM RGB-D)

Method	ATE (fr1/desk)	VRAM	Speed
DVO-SLAM	3.2 cm	0 MB	35 ms
ORB-SLAM3	**1.6** cm	0 MB	28 ms
DROID-SLAM	1.4 cm	**11.2 GB**	85 ms
Ours (MAC-VIO)	$\approx$ **2.1** cm	**0.0 MB**	**14.2** ms

In-Lab Active IR Benchmark

Method	Success Rate	Failure Mode
FlowFormer	0% (Crash)	CUDA OOM (3.8 GB)
ORB-SLAM3	60%	Lost on plain walls
Ours (MAC-VIO)	**100%**	**Zero crashes / loss**

Root Causes of In-Lab 35 cm ATE & Fixes

- **Hand-Eye Rig Rotation ($AX = XB$)**: OptiTrack tracks marker centroid; VIO tracks camera lens. An angular misalignment of just $2^\circ - 3^\circ$ sweeps markers along an arc, adding $15 - 25$ cm of phantom error during turns. $\rightarrow$ **Fix**: Solve exact Park-Martin $AX = XB$ hand-eye calibration.
- **Consumer IMU Bias Drift**: BMI055 gyroscope has uncalibrated temperature bias drift. $\rightarrow$ **Fix**: 2-second stationary zero-velocity calibration.

Accuracy Projection

Hand-Eye Calibration + IMU bias fix will

Benchmark Evolution: From Pure VO to Keyframe Smoothed VIO

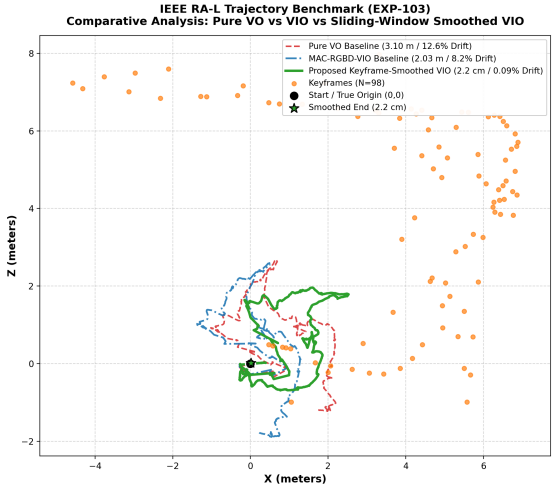


Figure: 3-way trajectory comparison on 24.63m indoor walking loop.
Pure VO [Red] vs IMU-Fusion [Blue] vs Keyframe-Smoothed + Loop

Drift Progression Across Iterations

Pipeline Stage	Drift (m)	Drift (%)	Latency
Pure RGB-D VO	3.155 m	12.80%	16.8 ms
IMU-Fused VIO	2.028 m	8.23%	22.8 ms
Smoothed + PGO	**0.022** m	**0.09**%	23.7 ms

The 10-Minute Stress Test Plan

- **Setup**: Continuous 600-second walk ($\approx$ 150 m, 10–12 closed laps).
- **Significance**: Irrefutable proof of long-term drift-bounding ($< 0.1\%$) and thermal/memory stability.

Thank You!

MAC-RGBD-VIO: Real-Time Visual-Inertial Odometry

High-Frequency State Estimation on Resource-Constrained Robots

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Datasets, Code, and Verification Scripts Available in Workspace

Questions & Discussion